ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MATHEMATICAL LITERACY

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mathematical Literacy paper 1 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct methods and sensible explanations.

QUESTION 1

1.1.1 $8\,500 + 560$ ✓ $= \text{R}9\,060$ ✓ (2)

1.1.2 $3\,200 + 2\,450 + 1\,320 + 600 + 250 + 400$ ✓ $= \text{R}8\,220$ ✓ (2)

1.1.3 Surplus ✓ $\text{of } 9\,060 - 8\,220 = \text{R}840$ ✓ (2)

1.1.4 $\frac{3\,200}{9\,060} \times 100$ ✓ $= 35{,}32\%$ ✓ (2)

1.1.5 Rent – fixed ✓; groceries – variable ✓ (2)

1.1.6 Any TWO: buy groceries in bulk / on special ✓; save electricity (switch off lights and geyser) ✓; use cheaper data bundles; walk or use cheaper transport where possible (2)

1.2 $300 \div 2{,}85$ ✓ $= 105{,}26\ \text{kWh}$ ✓ (2)

1.3.1 $1\% \times 12\,400$ ✓ $= \text{R}124$ ✓ (2)

1.3.2 $124 + 1\,150 + 650$ ✓ $= \text{R}1\,924$ ✓ (2)

1.3.3 $12\,400 - 1\,924$ ✓ $= \text{R}10\,476$ ✓ (2)

1.3.4 Gross salary is the total amount earned before deductions ✓; net salary is the amount received after deductions ✓ (2)

1.4.1 $480 \times 1{,}15$ ✓ $= \text{R}552$ ✓ (2)

1.4.2 $920 \div 1{,}15$ ✓ $= \text{R}800$ ✓ (2)

1.5 $1\,200 \times 0{,}75$ ✓ $= \text{R}900$ ✓ (2)

1.6 $\text{Interest} = 5\,000 \times 0{,}18 \times 1{,}5 = \text{R}1\,350$ ✓; $\text{total} = \text{R}6\,350$ ✓ (2)

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QUESTION 2

2.1 (a) 2 500 g ✓ (b) 0,75 ℓ ✓ (c) 180 cm ✓ (d) 225 minutes ✓ (4)

2.2.1 $2 \times (4 + 3{,}5)$ ✓ $= 15\ \text{m}$ ✓ (2)

2.2.2 $15 \times 2{,}6 = 39\ \text{m}^2$ ✓; $39 - 1{,}8 - 1{,}5$ ✓ $= 35{,}7\ \text{m}^2$ ✓ (3)

2.2.3 $35{,}7 \times 2 = 71{,}4\ \text{m}^2$ ✓; $71{,}4 \div 8$ ✓ $= 8{,}93\ \ell$ ✓ (3)

2.2.4 $8{,}93 \div 5 = 1{,}79 \rightarrow 2\ \text{tins}$ ✓✓; $2 \times \text{R}420 = \text{R}840$ ✓ (3)

2.3 $\text{06:45 to 08:10} = 1\ \text{hour } 25\ \text{minutes}$ ✓✓ (2)

2.4 $\frac{300}{4} \times 10$ ✓ $= 750\ \text{g}$ ✓ (2)

2.5 $(9 \div 5 \times 30) + 32$ ✓ $= 86\ °\text{F}$ ✓ (2)

2.6 $2\,500 \div 180 = 13{,}89$ ✓ → 13 full days ✓ (2)

2.7 $\text{Perimeter} = 2(12 + 8) = 40$ m ✓; $\text{cost} = 40 \times 65 = \text{R}2\,600$ ✓ (2)

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QUESTION 3

3.1.1 $7{,}4 \times 50\,000 = 370\,000$ cm ✓ $= 3\,700$ m ✓ $= 3{,}7$ km ✓ (3)

3.1.2 $\text{Time} = 3{,}7 \div 5 = 0{,}74$ h ✓ $= 44{,}4$ minutes ✓ (2)

3.1.3 1 unit (cm) on the map ✓ represents 50 000 of the same units in reality ✓ (2)

3.2.1 $4{,}5$ m ✓ by 3 m ✓ (2)

3.2.2 $4{,}5 \times 3$ ✓ $= 13{,}5\ \text{m}^2$ ✓ (2)

3.3 Any TWO: it stays correct when the map is enlarged or reduced ✓; distances can be measured directly without calculation ✓ (2)

3.4 South-west ✓✓ (2)

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QUESTION 4

4.1.1 Discrete ✓ – the hours were counted in whole numbers ✓ (accept continuous if the student argues that time can take any value, with that reason) (2)

4.1.2 3 hours ✓ (1)

4.1.3 $(1\times2) + (2\times5) + (3\times8) + (4\times7) + (5\times5) + (6\times3)$ ✓✓ $= 2 + 10 + 24 + 28 + 25 + 18 = 107$ ✓; $107 \div 30 = 3{,}57$ hours ✓ (4)

4.1.4 The 15th value is 3 and the 16th value is 4 ✓, so the median $= (3 + 4) \div 2 = 3{,}5$ hours ✓ (2)

4.1.5 $\frac{5+3}{30} \times 100$ ✓ $= 26{,}67\%$ ✓ (2)

4.1.6 Heading ✓; labelled axes ✓; correct bar heights ✓; equal widths and gaps ✓ (4)

4.2 Any TWO: how many users were asked ✓; who did the survey / was it independent ✓; how were the users chosen; what exactly was the question (2)

4.3 Advantage: easy to analyse / many people can answer quickly / anonymous ✓; disadvantage: friends are not representative of everybody, so the results are biased ✓✓ (3)

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QUESTION 5

5.1 (a) $\frac{6}{20} = \frac{3}{10}$ ✓ (b) $\frac{10}{20} = \frac{1}{2}$ ✓✓ (c) 0 ✓ (4)

5.2 Under similar conditions in the past, it rained on about 7 out of 10 such days ✓; rain is likely but not certain ✓ (2)

5.3 H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6 ✓✓ (12 outcomes); P(heads and even) $= \frac{3}{12}$ ✓ $= \frac{1}{4}$ ✓ (4)

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TOTAL: 100

END OF MARKING GUIDELINE